

Scalability & Future Plan

Date	28 September 2026
Team ID	SWTID-2026-7718
Project Name	AI Study Buddy API
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited support for large numbers of simultaneous users	May affect response time when many students use the system at the same time	High
2	Limited storage and handling of large study materials	Large files and increasing user data may require additional storage capacity	Medium
3	Limited personalization of learning activities	Study plans and recommendations may not be fully customized for every student	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed for normal student usage	Use scalable server resources and load balancing to support more users
2	Data Storage	Stores required user and application information	Increase database capacity and use cloud-based storage for study materials
3	Performance	Provides responses through the AI system	Optimize backend APIs and database queries to improve response time
4	Security	Basic user authentication	Implement stronger authentication, data protection, and secure access controls

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Voice-Based AI Interaction	3–4 Months	Allows students to interact with the AI using voice
Phase 2	Progress Tracking & Performance Analysis	5–6 Months	Helps students monitor their learning progress

Phase 3	Multilingual Support & Subject-Wise Recommendations	6–12 Months	Makes the system more accessible and provides personalized learning recommendations
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